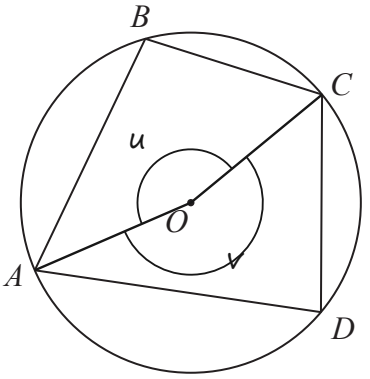
21 A, B, C and D are points on the circumference of a circle, centre O.



Prove that the sum of angle ABC and angle ADC is 180°

Draw radii AO and OC.

u is the central angle on arc ABC; v is the other.

u + v = 360° (angles around a point).

1 mark
Radii; angles around O

The angle at the centre is twice the angle at the circumference standing on the same arc.

1 mark
Centre-angle theorem

Angle ADC = $\frac{u}{2}$ Angle ABC = $\frac{v}{2}$

Angle ABC + angle ADC = $\frac{v + u}{2}$

1 mark
Add both angles

= $\frac{360}{2}$ = 180°.

Therefore angle ABC + angle ADC = 180°, as required.

1 mark
Complete proof with both reasons

(Total for Question 21 is 4 marks)